

LABORATORY OF AUTOMATION SYSTEMS

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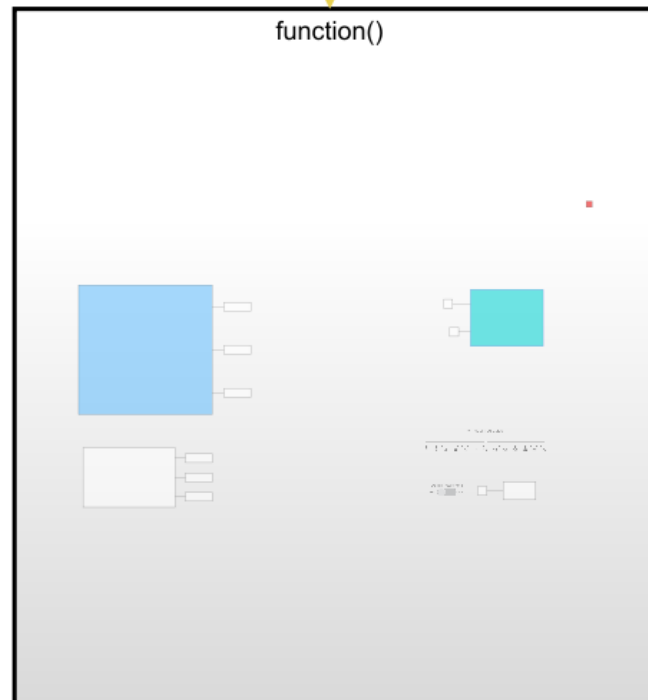
Email: davide.samori@unibo.it

You can download and open:

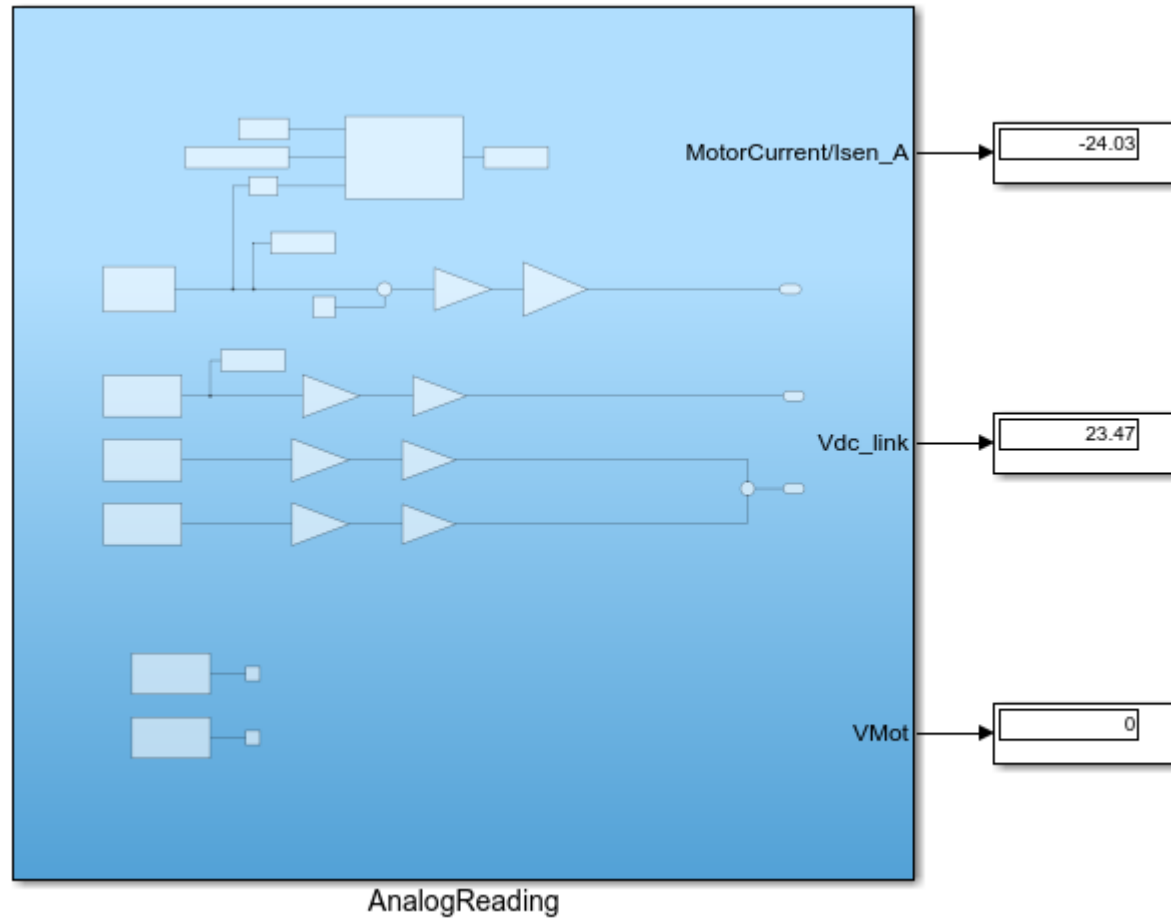
- BaseModelForControl.slx
- BaseModelForControlParameters.m
- Your DC Motor Model

Today Goals:

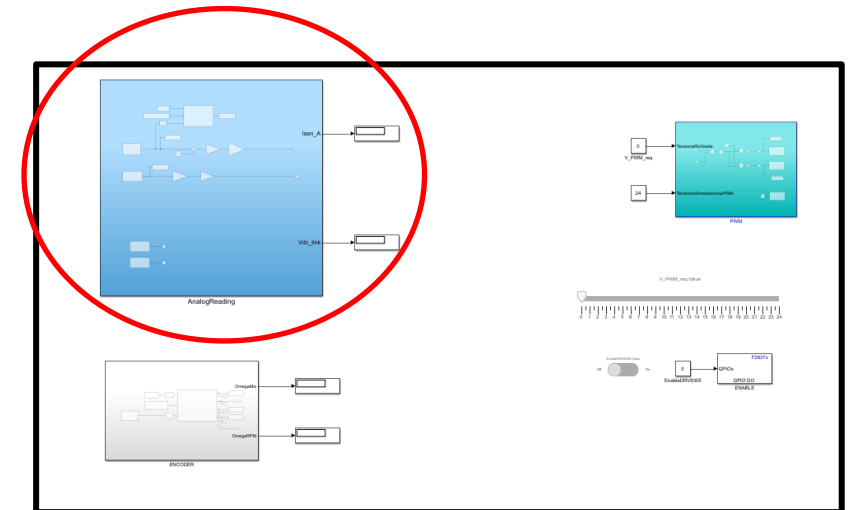
- Have a look of our first interface
- Check measure (I, V, Omega, ecc...)
- Some optimization
- Start PI design

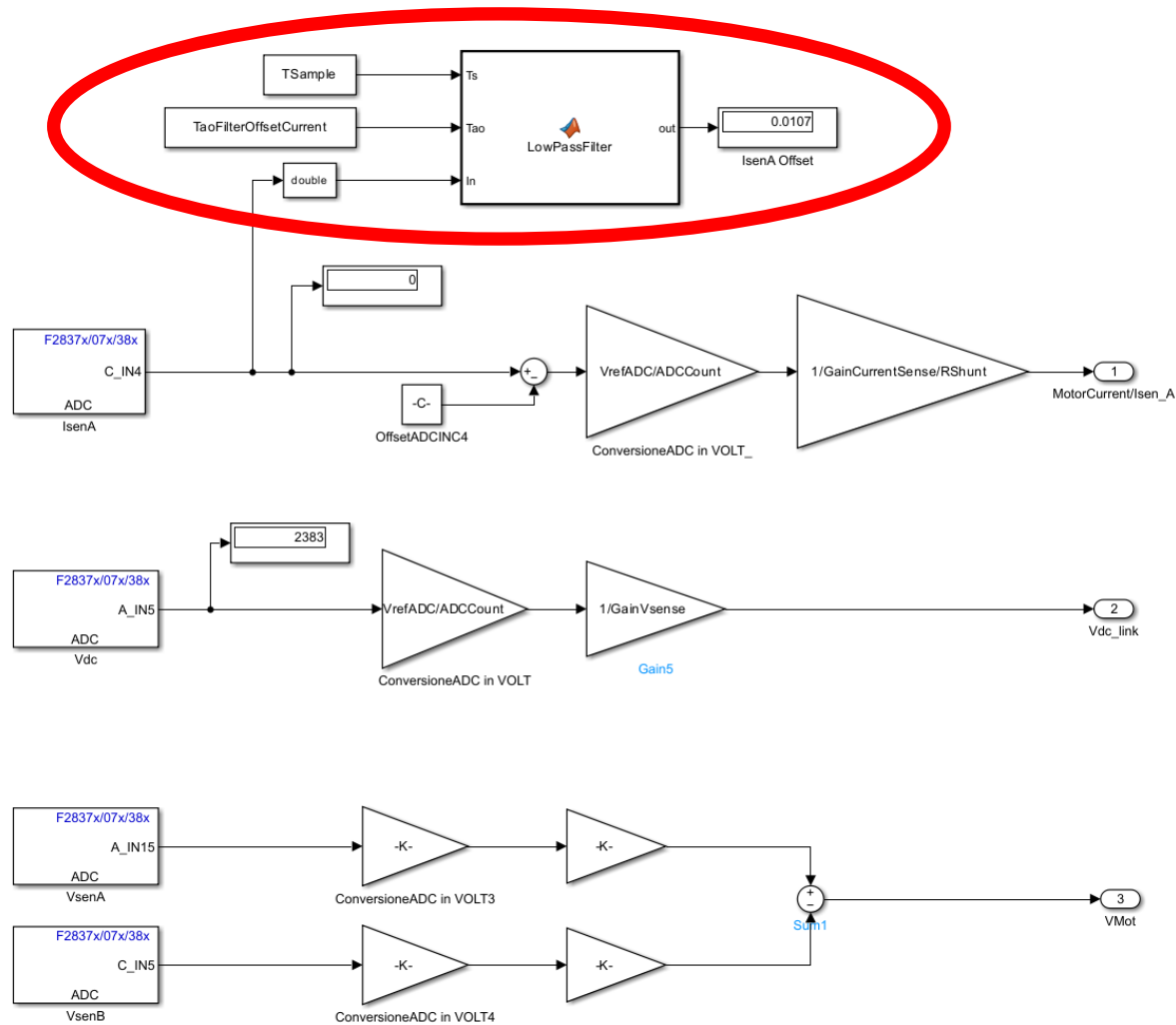


This is our main task.
Every TSample=100us we run it



Inside we have a block for analog reading

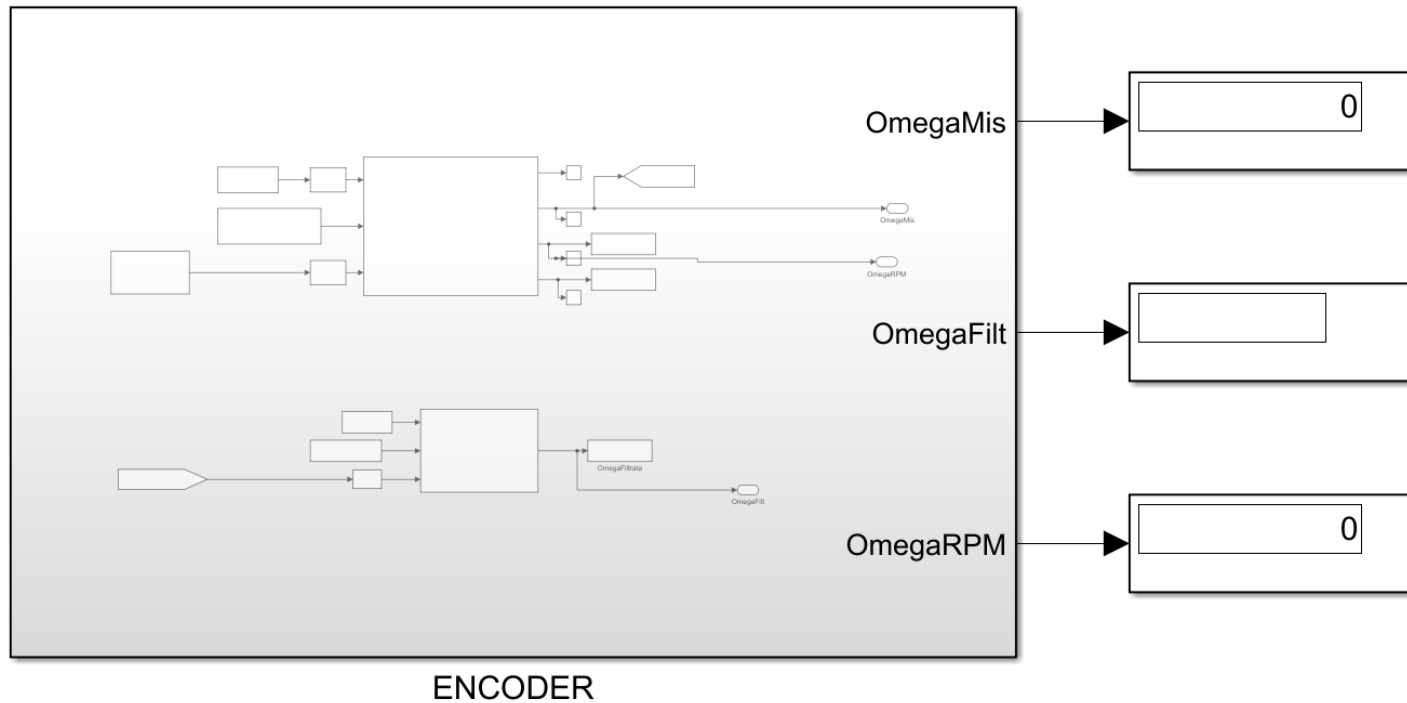




Motor Current

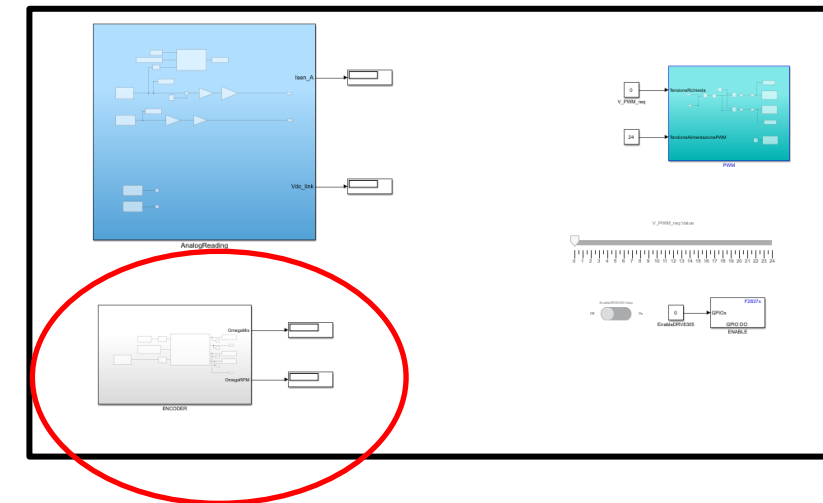
VDC_Link / PowerSupply

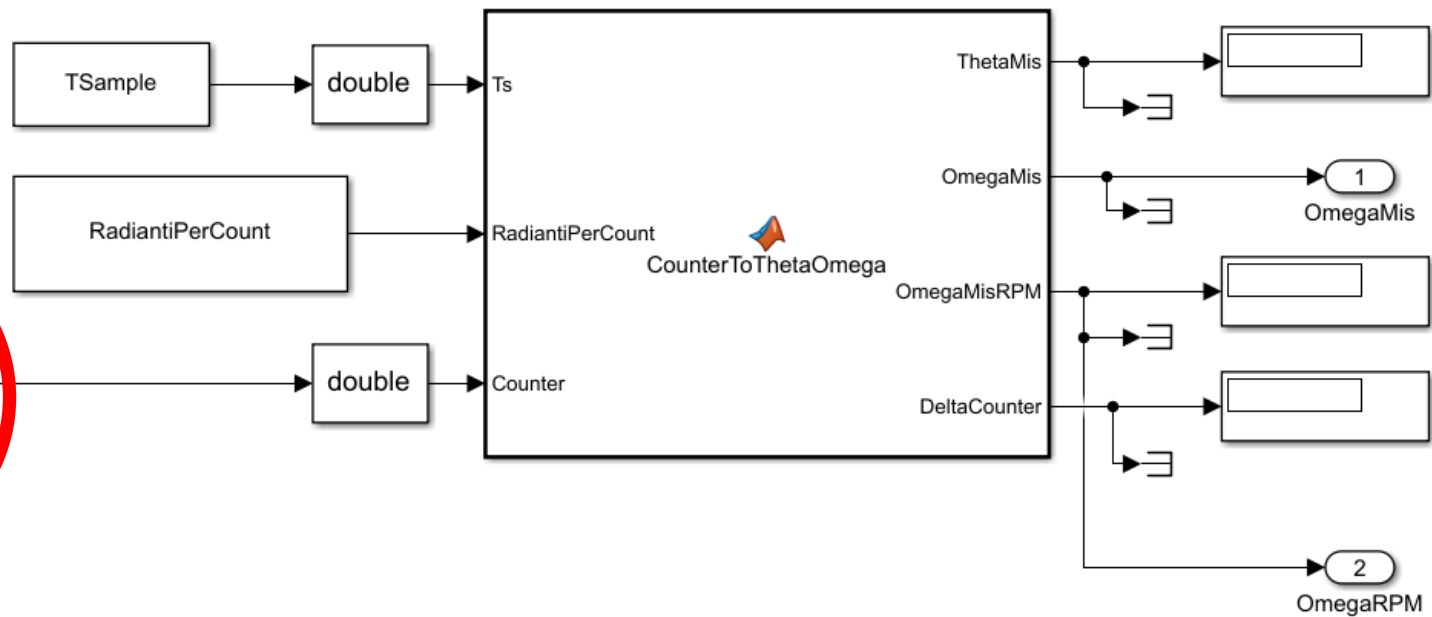
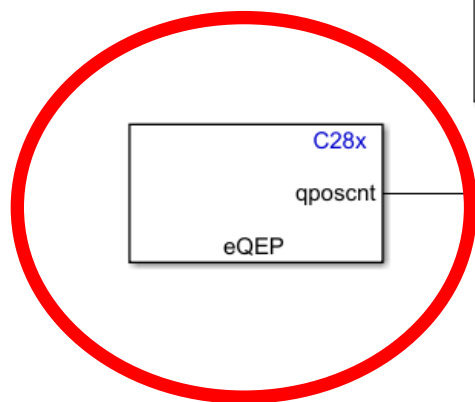
Motor Voltage

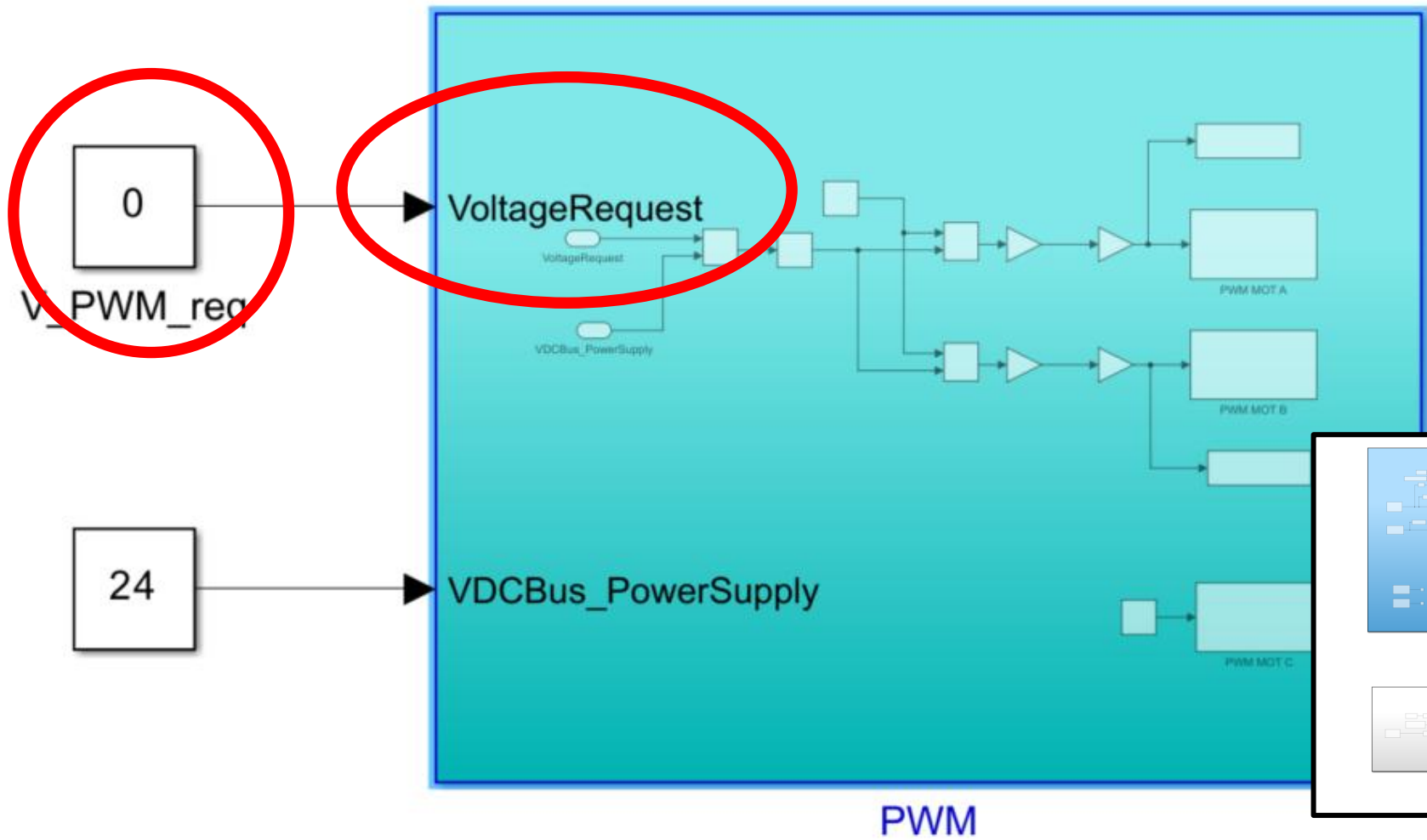


Encoder Block:

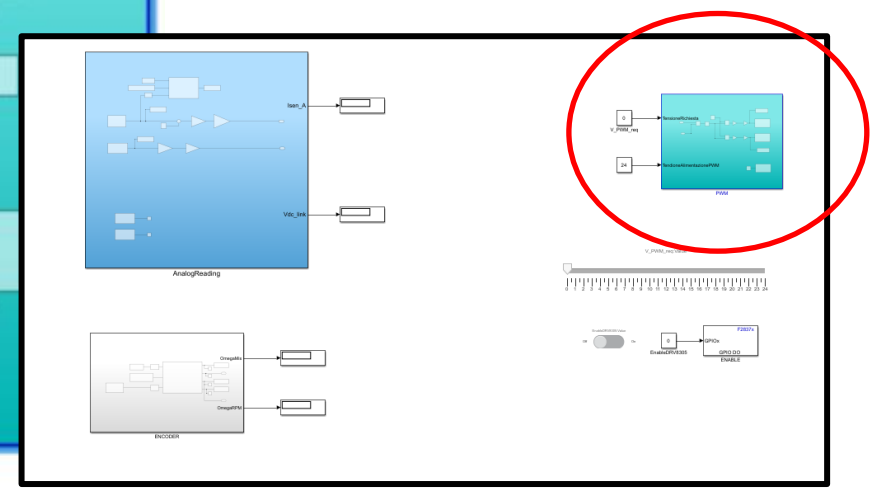
- OmegaMis (rad/s) S.I.
- OmegaFilt (rad/s) S.I.
- OmegaRPM (rpm)

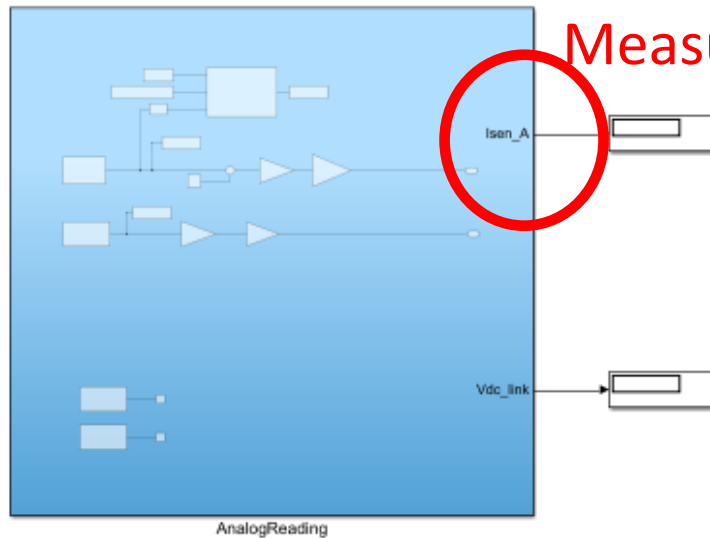






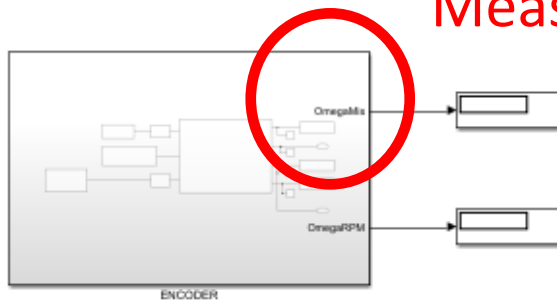
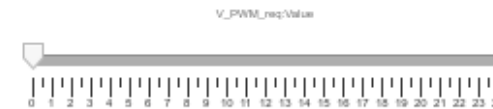
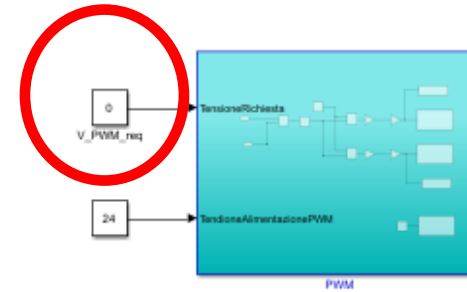
Here we can ask for a voltage: a value between -24V e +24V





Measured Current

Voltage to the motor



Measure speed (rad/s)



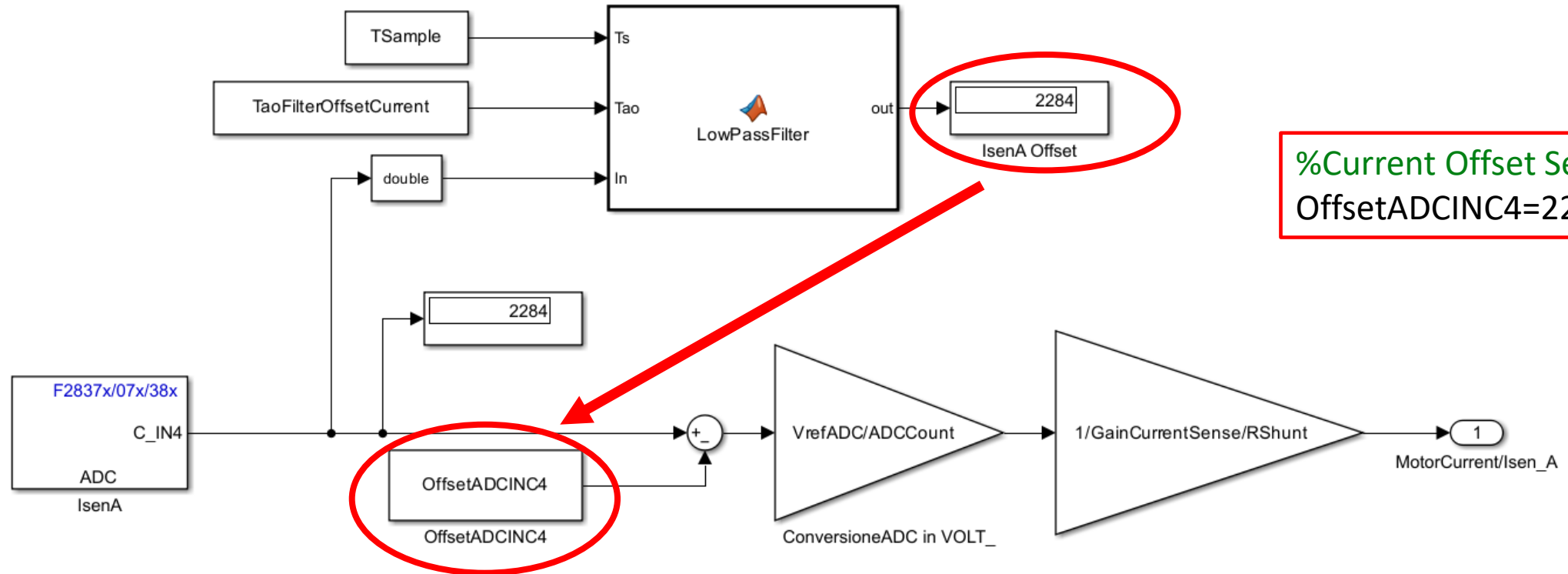
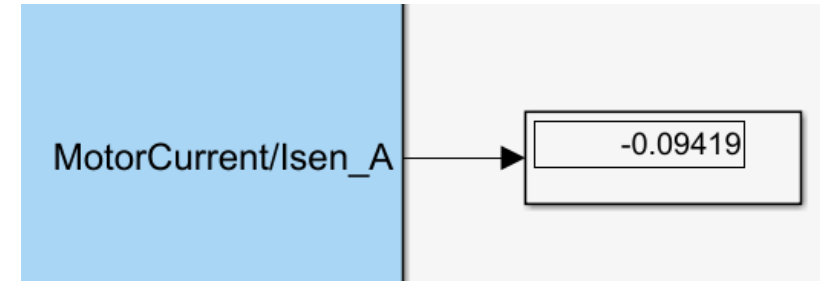
We don't have motor model, we use real motor

You can Run Model:

- Run the motor
- Watch Speed, Current, ecc...
- Current with/without Load

[illegible]

Voltage Request = 0V



%Current Offset Setup #2

```
OffsetADCINC4=2284;
```

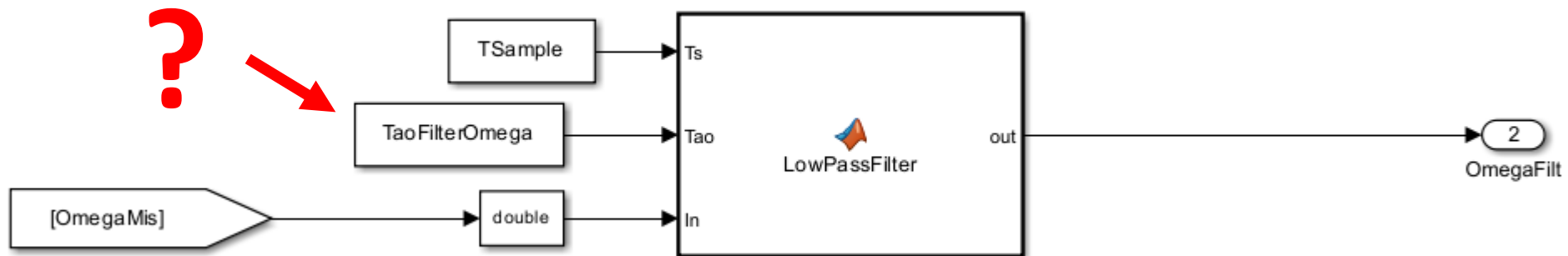
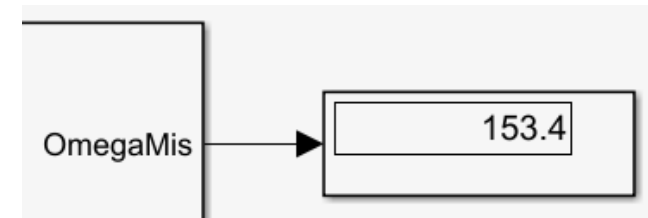
FILTER

Enable Drive = 1

Voltage Request = 24V

% Omega filter - To Be Defined

TaoFilterOmega=.....;



FF

Enable Drive = 1
Voltage Request = 12V



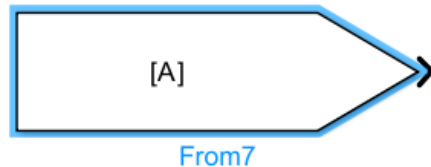
Change Power Supply Voltage
24V vs 15V



Look Vmotor and speed



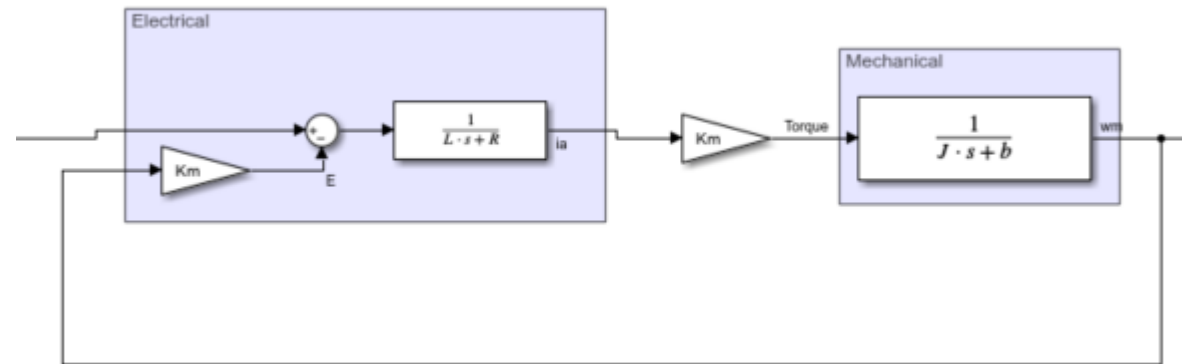
How can we avoid it quickly?



OK, now our interface runs well....

NOW, we want to start controlling our motor

Please open your DC Motor Model



Model of a DC electric motor

Note that the transfer function

$$G_{\omega}(s) = \frac{k_m}{L_A J s^2 + (R_A J + b L_A) s + (b R_A + k_m^2)}$$

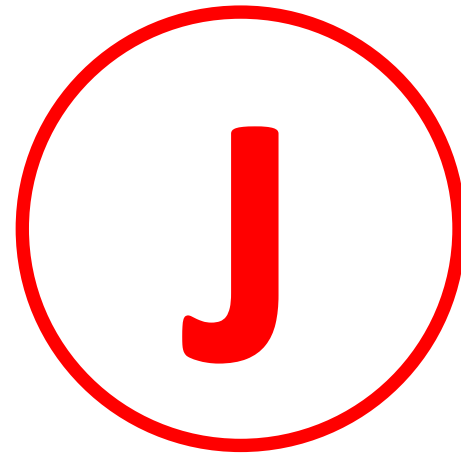
%% TRANSFER FUNCTION

s=tf('s');

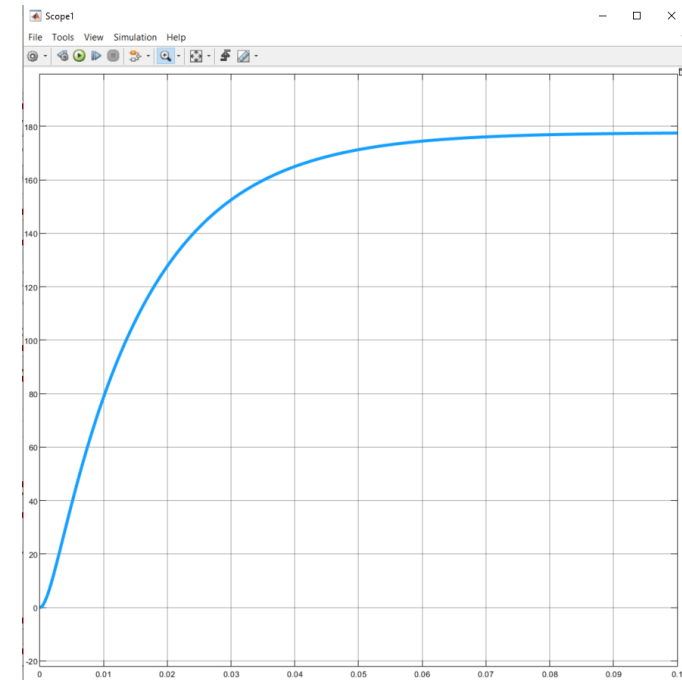
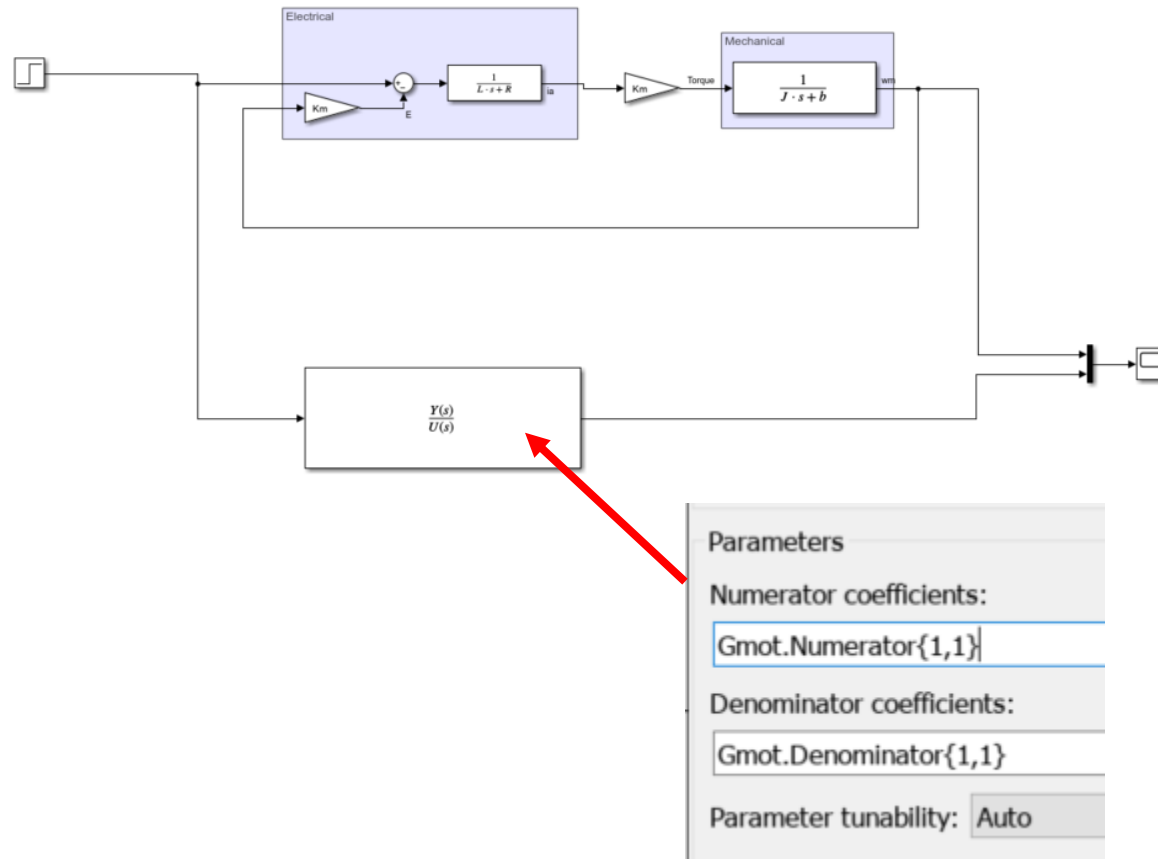
z=tf('z',Tsample);

%DC Motor continuous transfer function

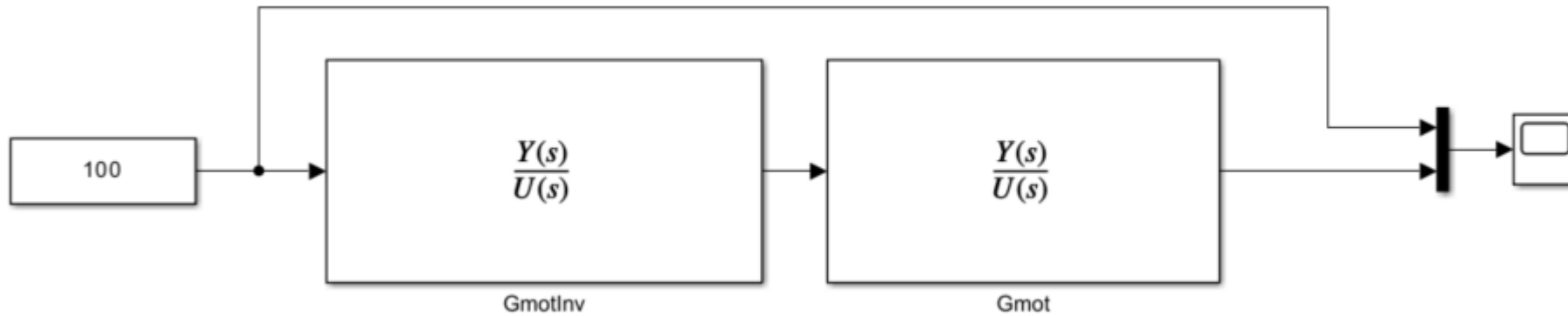
Gmot=tf([Km],[L*J R*J+b*L Km*Km+b*R])



A quick check...



$G_{motInv}=1/G_{mot}$



% I need two poles

pole1=1/(1+0.002*s);

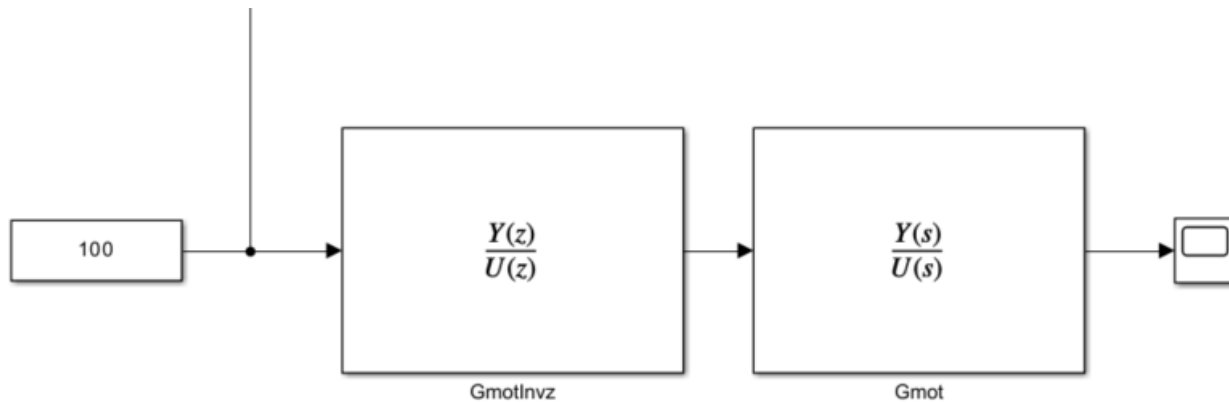
pole2=1/(1+0.001*s);

$G_{motInv}=G_{motInv}*pole1*pole1$

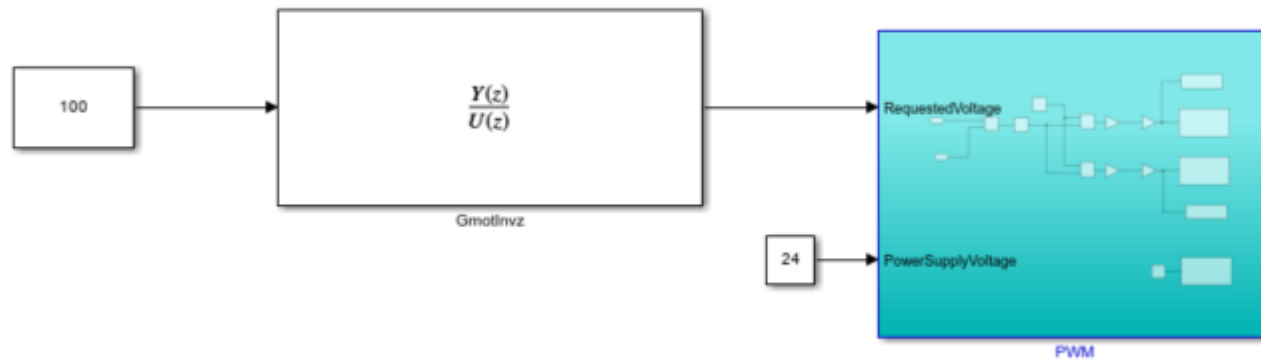


% GmotInv Discretization

GmotInvz=c2d(GmotInv,TSample)



You can copy this block to BaseModelForControl

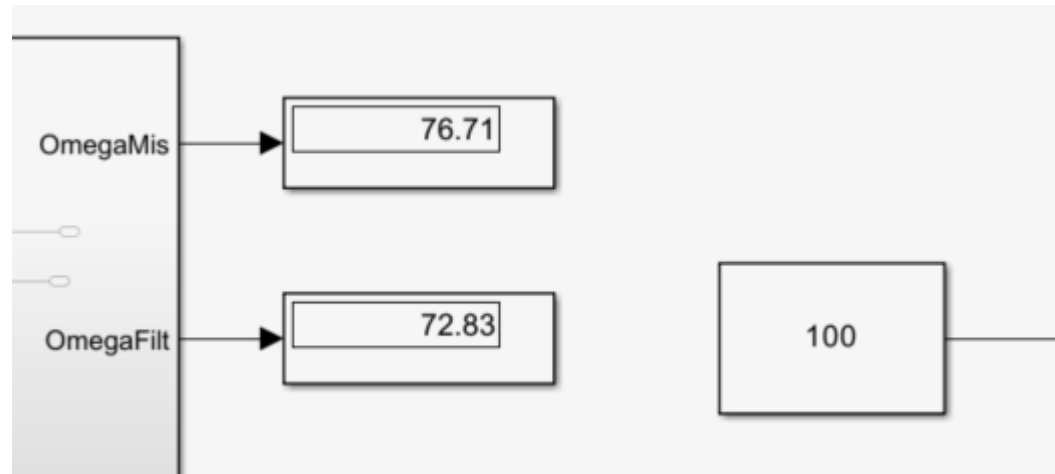


Did you have good result?



Try with brakes resistors...

What's happen?



Try with **$b = J/Taom$**

A quick review about the filter